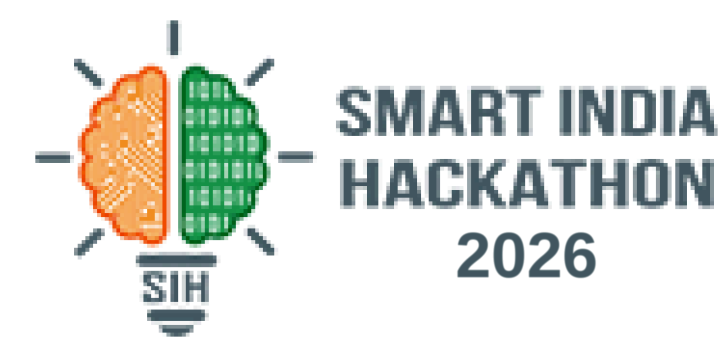
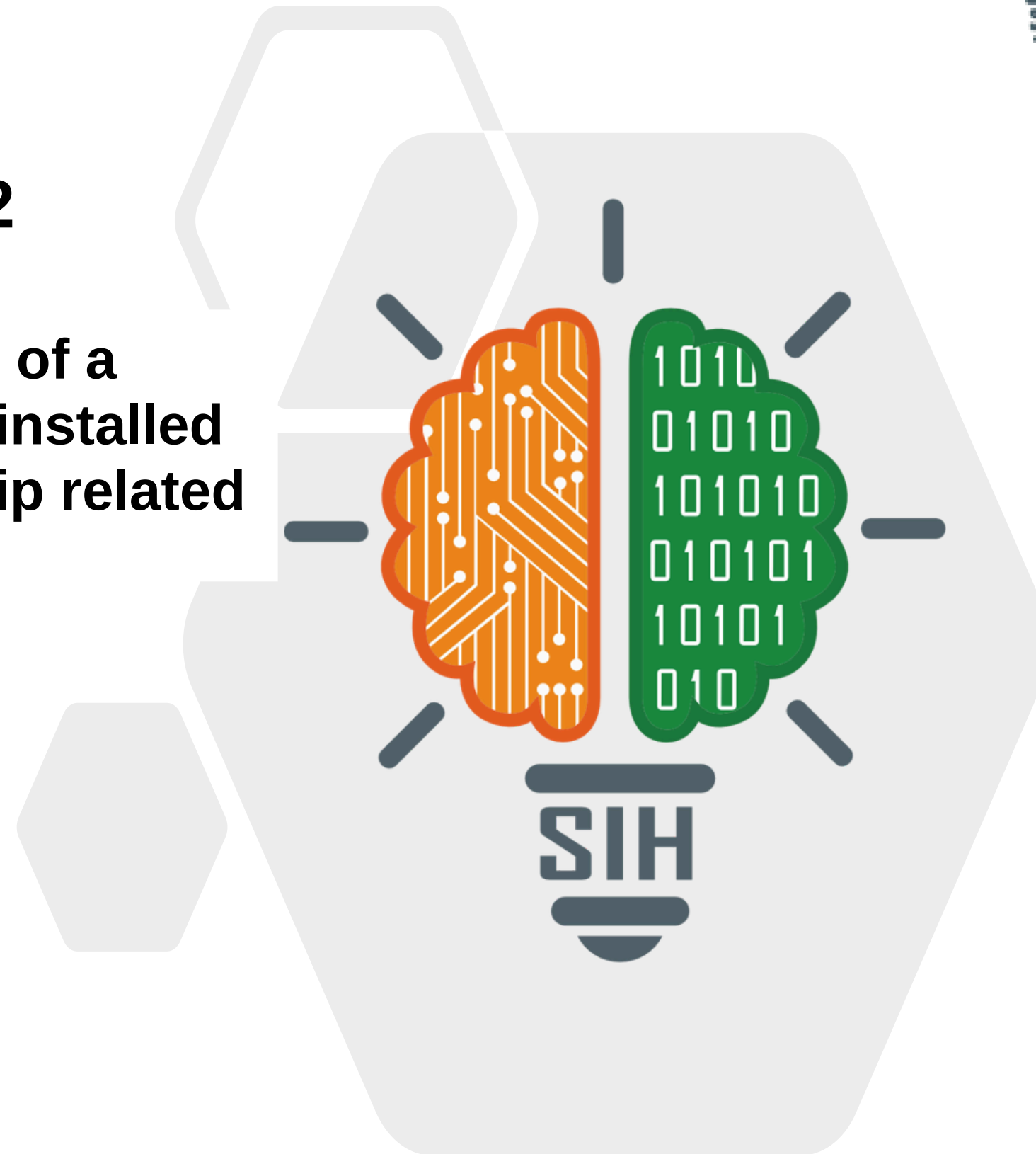


SMART INDIA HACKATHON 2026



- **Problem Statement ID - SIH25082**
- **Problem Statement Title - Development of a travel related software app that can be installed on mobile phones that could capture trip related information**
- **Theme - Travel & Tourism**
- **PS Category- Software**
- **Team ID-**
- **Team Name-Null Set**



NULL SET

PROPOSED SOLUTION



FEATURES



Problem

- Manual travel surveys are costly, time-consuming, and prone to errors.
- Low user participation results in incomplete mobility data.
- Existing solutions provide little value to end users.



Our Solution

- A smart mobile application that automatically captures travel information using smartphone sensors while delivering useful daily travel insights to users.

- Automatic Trip Detection
- AI-Based Transport Mode Classification
- Trip Chain Reconstruction
- Auto Fare & Fuel Tracking
- Monthly Travel Cost Calculator
- Alternate Route Suggestions
- Smart Trip Reminders & Local Travel Alerts
- Privacy Transparency Dashboard
- Eco-Points & Rewards System

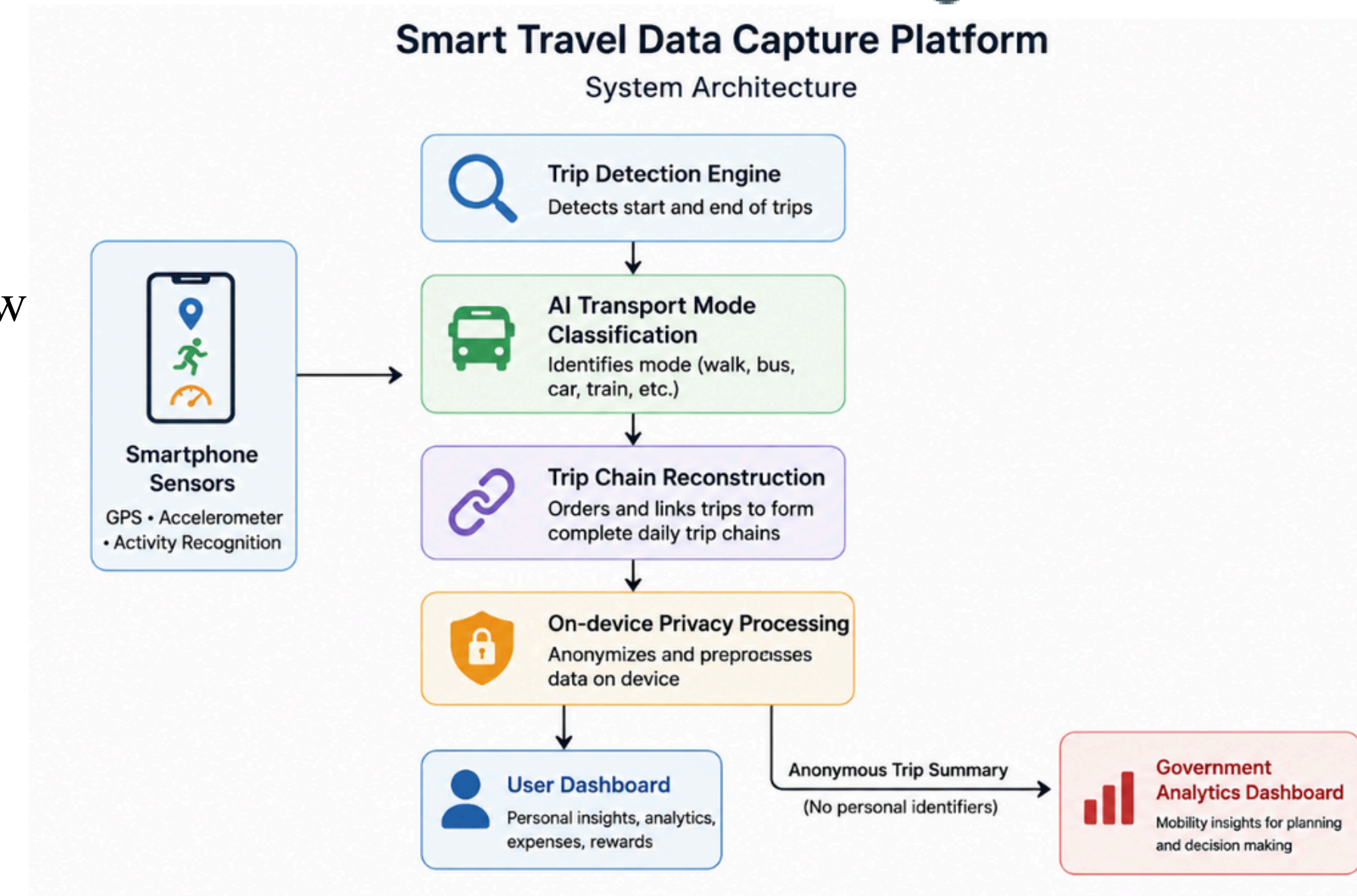
Government receives anonymized mobility insights while users receive a personal travel companion they actually want to use

Technologies to be used:

- **Frontend:** React Native, Google Maps SDK
- **Backend:** FastAPI, REST APIs
- **Database:** PostgreSQL, PostGIS
- **Machine Learning:** Activity Recognition API, TensorFlow Lite, GPS + Sensor Fusion
- **Core Workflow**
 1. Detect Trip Automatically
 2. Classify Transport Mode
 3. Build Trip Chain
 4. Generate User Insights
 5. Anonymize Data
 6. Share Mobility Analytics

Methodology and process of implementation

- **Agile Development:** Sprint-based development with continuous testing and improvements.
- **AI & Sensor Fusion:** Automatic trip detection and transport mode classification using smartphone sensors.
- **Privacy-First Design:** On-device processing with anonymized government data sharing.





• Analysis of the feasibility of the idea:

- Built using React Native, FastAPI, PostgreSQL & AI modules → Technically feasible.
- Smartphone sensors eliminate additional hardware → Cost-effective solution.
- Privacy-first architecture supports secure government deployment → Operationally viable.



• Potential challenges and risks:

- GPS inaccuracies & battery consumption during continuous tracking.
- AI prediction errors in transport mode classification.
- User privacy concerns and permission dependency.



• Strategies for overcoming these challenges:

- Sensor fusion & confidence-based AI for improved accuracy.
- Adaptive GPS sampling to optimize battery usage.
- On-device processing & anonymized data sharing for privacy.

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IMPACT AND BENEFITS



- **Potential impact on the target audience**



- **Government:** Real-time mobility insights for data-driven planning.



- **Citizens:** Smart travel assistant with expense tracking & travel insights.



- **Transport Agencies:** Automated surveys & reduced operational costs.

- **Benefits of the solution (social, economic, environmental, etc.)**



- **Social:** Better public transport & commuter experience.



- **Economic:** Lower survey costs & travel expenses.



- **Environmental:** Reduced congestion & carbon emissions.



- **Technological:** AI-powered, privacy-first mobility platform.

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RESEARCH AND REFERENCES



- Details / Links of the reference and research work

-  • Android Activity Recognition API – Google Developers
<https://developers.google.com/location-context/activity-recognition>
-  • Google Maps Platform Documentation
<https://developers.google.com/maps>
-  • OpenStreetMap Documentation
<https://www.openstreetmap.org/>
-  • ResearchGate – Smartphone-based Travel Behaviour Analysis
<https://www.researchgate.net/>
-  • TensorFlow Lite Documentation (On-device Machine Learning)
<https://www.tensorflow.org/lite>